Factors Influencing Agricultural Systems in an Agribusiness Company

1. Introduction

Agriculture is an important sector that provides food, raw materials, employment, and income to millions of people. Modern agriculture is no longer dependent only on traditional farming methods. It is influenced by several factors such as weather conditions, technology, government policies, economic conditions, societal trends, market competition, and customer needs.

An agribusiness company must understand these factors before making decisions about crop selection, irrigation, machinery, labour, production, storage, marketing, and distribution. Effective farm management helps the company increase productivity, reduce costs, use resources efficiently, and satisfy customers.

2. Background of the Agribusiness Company

GreenGrow Agro Pvt. Ltd. is an agribusiness company involved in the cultivation, processing, packaging, and marketing of vegetables and fruits. The company operates several farms and supplies its products to supermarkets, local markets, restaurants, and food-processing companies..

Although the company was producing good-quality crops, it faced several challenges:

- Unpredictable weather conditions
- Increasing production costs
- Water shortages
- Rising labour costs
- Changing customer preferences

3. Problem Statement

The major problem faced by GreenGrow Agro was that agricultural decisions were not sufficiently based on changing environmental, technological, economic, legal, and market conditions.

For example, the company sometimes produced large quantities of a crop when market demand was low. In other situations, unexpected rainfall affected crop quality. Rising fertilizer and labour costs also reduced profit.

4. Factors Influencing Agricultural Systems

4.1 Weather and Climate

Agricultural production depends heavily on:

- Rainfall
- Temperature
- Humidity
- Wind
- Sunlight
- Drought

Management Response

The company introduced better weather monitoring. Farm managers began using weather forecasts to plan:

- Irrigation
- Fertilizer application
- Spraying
- Planting
- Harvesting
- Pest management

Result

Better weather-based planning reduced crop losses and improved water management.

5. Technology

Technology has transformed modern agriculture. Agricultural technology can help farmers increase productivity while reducing resource consumption.

Important agricultural technologies include:

- Precision agriculture
- Drones
- GPS-based machinery
- Soil sensors
- Weather sensors
- Automated irrigation
- Satellite imagery

Benefits

Technology helped the company:

- Reduce water wastage
- Improve crop monitoring
- Reduce unnecessary input use
- Detect problems earlier
- Improve productivity

6. Legal Policies and Government Regulations

Agricultural businesses must operate according to government laws and regulations. These may relate to:

- Land use
- Water usage
- Environmental protection
- Fertilizer use
- Pesticide application.

7. Economic Factors

Economic conditions have a major influence on agricultural systems.

Important economic factors include:

- Cost of seeds
- Fertilizer prices
- Labour costs
- Fuel prices
- Machinery costs
- Electricity costs
- Transportation costs

8. Societal Trends

Society's attitudes and lifestyles influence agricultural production and consumption.

Modern consumers are increasingly interested in:

- Healthy food
- Food safety
- Environmentally friendly farming

- Organic products
- Locally produced food
- Sustainable packaging

9. Market Competition

Agricultural businesses operate in competitive markets. Competition can come from:

- Local farmers
- Large agricultural companies
- Cooperatives
- Food-processing companies
- Imported products
- Online agricultural marketplaces

Companies must maintain good quality while controlling costs.

Situation at GreenGrow Agro

GreenGrow faced competition from companies offering similar vegetables at lower prices.

Instead of reducing prices alone, GreenGrow focused on differentiation.

The company improved:

- Product quality
- Packaging
- Supply reliability
- Customer service
- Product traceability
- Sustainable farming practices

10. Customer Needs

Customer needs are one of the most important factors influencing agricultural production.

Customers may consider:

- Price
- Quality
- Freshness
- Safety

- Appearance

11. Integrated Farm Management Approach

After studying these factors, GreenGrow Agro developed an integrated farm management system.

Step 1 – Collect Information

The company collected information about weather, soil, market demand, input costs, government regulations, and customer preferences.

Step 2 – Analyze the Information

Farm managers analyzed the collected information to identify risks and opportunities.

Step 3 – Select Crops

Crop selection was based on climate suitability, expected demand, water availability, and profitability.

Step 4 – Plan Resources

The company planned the required:

- Seeds
- Fertilizers
- Water
- Machinery
- Labour

Step 5 – Use Technology

Sensors, digital records, weather information, and data analysis were used to support farm decisions.

Step 6 – Monitor Production

The company regularly monitored crop health, resource use, production costs, and market conditions.

Step 7 – Evaluate Results

After harvesting, the company compared actual production and profits with planned results.